

Genetic report for longevity

Migraine and Headaches

Syringe (vascular and neuronal), headache in wounds, persistent headaches, triggers and drug response

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Migraine is very hereditary (about 40–50%) and polygenic: loci are distributed between the endothelium and vascular smooth muscle (PHACTR1, LRP1, HPSE2, FHL5) and neurons and ionic channels (TRPM8, KCNK5, MEF2D, PRDM16). This report reads these loci, MTHFR (migraine with aura), three polygenic scores of migraine and persistent headaches, the loci of headache in saved and some variants of drug response and caffeine. It is useful mainly because several drugs that already take (tadalafil, isotretinoin) can give headaches.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

26

variants

19

in the chip

6

imputed
($DR2 \geq 0.8$)

1

not found

6

Scores with
percentile

Overall reading. The genetic predisposition for migraine is on the lower side of the typical: the two migraine scores are at the 29 and 32 percentiles, and that of headaches for more than 3 months in the 40. In loci, there is a mixture: in favor, FHL5 AA, KCNK5 CC and SLC24A3 GG (without risk alleles); against, PHACTR1 AA, PRDM16 CC, ITPK1 AA and PLCE1 GG (2 risk alleles each) and NRP1 with 1. PHACTR1 AA is the allele that protects from coronary arteries, so "against" here is the other face of a vascular force. **MTHFR** 677 is heterozygous (G/A): the risk of migraine with aura rises mainly in homozygous, and the 5-MTHF of the plan already covers methylation. The headache is in the middle (the 62nd percentile, only 4 loci). The self-reported headache score (including tension) is at the 79th percentile, but it is an unweighted and unreliable allele count. In practice, genes do not ask for prevention; what can give more headaches are drugs (tadalafil, isotretinoin, retatrutide, dapagliflozin) and dehydration.

Favourable

- **FHL5 rs67338227 AA:** FHL5/UFL1: vascular smooth muscle.
- **KCNK5 rs10456100 CC:** KCNK5: potassium channel of two pores.
- **SLC24A3 rs4814864 GG:** SLC24A3: calcium exchanger.

Unfavourable

- **Headache (self-reported, including tension):** the 79th percentile
- **PHACTR1 rs9349379 AA:** PHACTR1 (EDN1): the allele that protects from migraine increases coronary risk, and vice versa.
- **PRDM16 rs2651899 CC:** PRDM16: one of the first loci of migraine.
- **NRP1 rs2506142 AG:** NRP1: neuropilin 1.
- **ITPK1 rs11624776 AA:** ITPK1.
- **PLCE1 rs2274224 GG:** PLCE1: phospholipase C epsilon.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 19 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 1 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Migraine: vascular component

Loci expressed in the endothelium and smooth muscle of the vessels. Several are shared with coronary disease and arterial dissection (PHACTR1, LRP1), often in the opposite direction.

Ageing mechanisms: Vascular Aging; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Migraine (recent multiloci score)	Typical / intermediate	the 32nd percentile · z=-0.45	Higher score : higher risk. PGS004797 (PGS Catalog; PRSmix; Migraine; PGP000604, PMID 38508198). Variants with genotype: 19 of 23 (covering of 93.1% coefficients).

Favourable

- FHL5 rs67338227 AA

Unfavourable

- PHACTR1 rs9349379 AA
- PLCE1 rs2274224 GG

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FHL5	rs67338227 · 6:96594271 · A/C/G/T	AA	chip	T (0; —)	strong	FHL5/UFL1: vascular smooth muscle. Meaning: GWAS Catalog: Migraine (Gormley P 2016; p=2e-27); T ↑ allele

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PHACTR1	rs9349379 · 6:12903725 · A/C/G/T	AA	chip	A (2; 60%)	strong	PHACTR1 (EDN1): the allele that protects from migraine increases coronary risk, and vice versa. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=1e-47); Allele A ↑
PLCE1	rs2274224 · 10:94279840 · G/A/C/T	GG	imputation · DR2=1.00	G (2; 54%)	moderate	PLCE1: phospholipase C epsilon. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=3e-26); G ↑ allele

Typical / intermediate variants (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LRP1	rs11172113 · 12:57133500 · T/C/G	CT	chip	T (1; 62%)	strong	LRP1: lipoprotein receptor and vascular signaling; one of the strongest loci. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=1e-90); allele T ↑
HPSE2	rs12260159 · 10:98942980 · G/A/C	GG	imputation · DR2=1.00	G (2; 91%)	moderate	HPSE2: heparanase 2. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=7e-16); G ↑ allele
MTHFR	rs1801133 · 1:11796321 · G/A/C	GA	chip	A (1; 36%)	moderate	C677T: T allele (A in strand) is associated with migraine in meta-analyses, especially in homozygous.
ZCCHC14	rs8052831 · 16:87544433 · A/G/T	AG	imputation · DR2=1.00	G (1; 36%)	moderate	ZCCHC14. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=8e-15); G ↑ allele

ADAMTSL4	rs6693567 1:150538184 C/T	TC	chip	C (1; 26%)	moderate	ADAMTSL4: extracellular matrix. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=1e-13); allele C ↑
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Migraine: neuronal component and ionic channels

TRPM8 (cold sensor), KCNK5 (potassium channel), MEF2D and PRDM16 affect the excitability of the trigemium and cortex. Familial hemiplegic migraine (CACNA1A, ATP1A2, SCN1A) is monogenic and rare; it is not read by chip.

Ageing mechanisms: Neuroinflammation; Mitochondrial Calcium Deregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Migraine (329 variants)	Typical / intermediate	the 29th percentile · z=-0.56	Higher score : higher risk. PGS001282 (PGS Catalog; snpnet; Migration (time-to-event); PGP000244, PMID 35324888). Variants with genotype: 288 of 329 (covering of 91.7% coefficients).
Migraine without aura	Typical / intermediate	the 48th percentile · z=-0.08	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 8 of 9 loci with genotype. Studies: Chen SP (Migraine without aura); Gormley P (Migraine without aura); Lin GY (Restless leg syndrome in migraine without aura). Variants with genotype: 8 of 9 (covering of 88.9% coefficients).

Favourable

- KCNK5 rs10456100 CC
- SLC24A3 rs4814864 GG

Unfavourable

- PRDM16 rs2651899 CC
- NRP1 rs2506142 AG
- ITPK1 rs11624776 AA

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
KCNK5	rs10456100 6:39215694 · C/T	CC	chip	T (0; 26%)	moderate	KCNK5: potassium channel of two pores. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=9e-19); allele T ↑
SLC24A3	rs4814864 20:19489173 · G/A/C/T	GG	chip	C (0; 26%)	moderate	SLC24A3: calcium exchanger. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=1e-28); allele C ↑

Unfavourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PRDM16	rs2651899 1:3167148 · T/A/C/G	CC	chip	C (2; 42%)	strong	PRDM16: one of the first loci of migraine. Meaning: GWAS Catalog: Migraine (Antila V 2013; p=4e-14); allele C ↑
NRP1	rs2506142 10:33179196 · A/G	AG	chip	G (1; 20%)	moderate	NRP1: neuropilin 1. Meaning: GWAS Catalog: Migraine (Gormley P 2016; p=2e-09); G ↑ allele
ITPK1	rs11624776 14:93129246 · A/C/T	AA	chip	A (2; 65%)	moderate	ITPK1. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=1e-18); Allele A ↑

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TRPM8		TT	chip	T (2; 83%)	strong	TRPM8: sensor of cold and menthol in sensory neurons. Meaning: GWAS

	rs10166942 2:233916448 · T/A/C					Catalog: Migraine (Hautakangas H 2022; p=9e-51); allele T ↑
TSPAN2	rs2078371 1:115134562 · T/A/C/G	TT	imputation DR2=0.98	C (0; 11%)	moderate	TSPAN2: oligodendrocytes. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=6e-42); allele C ↑
ASTN2	rs6478241 9:116490350 · A/G/T	GA	chip	A (1; 38%)	moderate	ASTN2: neuronal migration. Meaning: GWAS Catalog: Migraine (Gormley P 2016; p=1e-12); Allele A ↑
CFDP1	rs8046696 16:75408245 · T/A/G	GT	chip	T (1; 44%)	moderate	CFDP1/BCAR1. Meaning: GWAS Catalog: Migraine (Hautakangas H 2022; p=5e-14); T ↑ chip allele per position (ID 16:75442143)

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MEF2D	rs1050316 1:156464911 · G/ A/C/T	GT	chip	—	moderate	MEF2D: neural transcription factor.

Sustained headache and headache

Sage headache has a very marked circadian rhythm and few known loci (DUSP10, MERTK, FTCDNL1, LINC01877). Persistent headaches (more than 3 months) include chronic migraine and tension headache.

Ageing mechanisms: Circadian disorders; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Headaches for more than 3 months	Typical / intermediate	the 40th percentile · z=-0.25	Higher score : higher risk of persistent headache. PGS001928 (PGS Catalog; Penalized regression (bigstatsr); Headaches for 3+ months; PGP000263, PMID 34995502). Variants with genotype: 5549 of 5709 (covering of 97.0% coefficients).
Headache in saves	Typical / intermediate	the 64th percentile · z=0.32	Higher score : higher risk. Few loci. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 4 of 4 loci with genotype. Studies: O'Connor E (Cluster headache). Variants with genotype: 4 of 4 (covering of 100.0% coefficients).
Headache (self-reported, including tension)	Unfavourable	the 79th percentile · z=0.85	Higher score = more headaches. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 75 of 89 loci with genotype. Studies: Islam MR (Headache or HOMA-B); Islam MR (Headache or HOMA-IR); Islam MR (Headache or HbA1c levels); Islam MR (Headache or fasting glucose levels); Islam MR (Headache or fasting insulin levels); Islam MR (Headache or glucose levels); Islam MR (Headache or type 1 diabetes); Islam MR (Headache or type 2 diabetes). Variants with genotype: 75 of 89 (covering of 84.3% coefficients).

Favourable

None.

Unfavourable

- Headache (self-reported, including tension) (the 79th percentile)

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LINC01877	rs113658130 · 2:199639486 · C/T	TC	imputation · DR2=0.92	C (1; 66%)	moderate	LINC01877: locus of headache in rescues. Meaning: GWAS Catalog; Cluster headache (O'Connor E 2021; p=2e-17); C ↑ allele

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DUSP10	rs11579212 · 1:221899477 · A/C	AC	chip	—	moderate	DUSP10: locus of headache in rescues.
MERTK	rs10184573 · 2:199583530 · G/T	GT	imputation · DR2=0.94	—	moderate	MERTK: locus of headache in rescues.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FTCDNL1	— · —	—	SNP not found in the data	—	moderate	FTCDNL1: locus of headache in rescues. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Triggers and drug response

Methylation (MTHFR), caffeine (CYP1A2, ADORA2A) and tryptan response (GNB3) are the most studied associations, all of which have weak to moderate evidence.

Ageing mechanisms: Deregulation of Nutritional Sensing; Sleep disturbance

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP1A2	rs762551 15:74749576 · C/A/G	AC	chip	—	weak	CYP1A2 *1F: caffeine is acute treatment and, in excess or in abstinence, trigger. Information (profile: A/C, slower metabolization).
ADORA2A	rs5751876 22:24441333 · T/C	CC	chip	—	weak	ADORA2A 1976T>C: caffeine sensitivity. Information.
GNB3	rs5443 12:6845711 · C/T	CT	chip	—	weak	GNB3 C825T: the T allele was associated with the best response to triptans in headaches. Information.
COMT	rs4680 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met: inconsistent associations with migraine. Information.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Neuroinflammation	PRDM16 CC; NRP1 AG; ITPK1 AA; Headache (self-reported, including tension) (the 79th percentile)	KCNK5 CC; SLC24A3 GG	Migraine (329 variants) (the 29th percentile); Migraine without aura (the 48th percentile); TRPM8 TT; TSPAN2 TT; ASTN2 GA; CFDP1 GT; Headaches for more than 3 months (the 40th percentile); Headache in saves (the 64th percentile)
Mitochondrial Calcium Deregulation	PRDM16 CC; NRP1 AG; ITPK1 AA	KCNK5 CC; SLC24A3 GG	Migraine (329 variants) (the 29th percentile); Migraine without aura (the 48th percentile); TRPM8 TT; TSPAN2 TT; ASTN2 GA; CFDP1 GT
Intercellular Communication	PHACTR1 AA; PLCE1 GG	FHL5 AA	Migraine (recent multiloci score) (the 32nd percentile); LRP1 CT; HPSE2 GG; MTHFR GA; ZCCHC14 AG; ADAMTSL4 TC
Vascular Aging	PHACTR1 AA; PLCE1 GG	FHL5 AA	Migraine (recent multiloci score) (the 32nd percentile); LRP1 CT; HPSE2 GG; MTHFR GA; ZCCHC14 AG; ADAMTSL4 TC
Circadian disorders	Headache (self-reported, including tension) (the 79th percentile)	—	Headaches for more than 3 months (the 40th percentile); Headache in saved (the 64th percentile); LINC01877 TC
Nutrient Sensing Deregulation	—	—	—
Sleep disturbance	—	—	—

Practical part

Ideas for headache, crossed with the genetic plan v2 and current medication. There is no known complaint of migraine: none of this asks to change the stack, everything is suggestion (not applied) in case headaches appear. Magnesium (glycinate) and fish oil that already takes are among the prevention options with some evidence, and telmisartan is of the same class as candesartan, used in the prevention of migraine.

1. Tadalafil: headache is the most common adverse effect

Genetics: Vascular loci (PHACTR1, LRP1, PRDM16)

PDE5 inhibitors give headache in 10–15% of people (via NO-cGMP, the same one that triggers migraine). If headache appears in the following hours of taking, talk to the doctor about lowering the daily dose.

V2 genetic plan and medication: Compatible with the plan (tadalafil remains). • *suggestion (not applied)*

2. Isotretinoin: never with tetracyclines

Genetics: No direct genetic relationship

Isotretinoin can give headache and, rarely, intracranial hypertension; the risk rises greatly with doxycycline or minocycline. Strong and persistent headache, in the morning, with blurred vision or nausea, requires rapid medical evaluation.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Hydration and regular meals

Genetics: Persistent headache score p40

Dapagliflozin (diuresis) and retatrutide (less hunger, less liquid) favor dehydration and long fasts, two classic triggers. Water throughout the day, electrolytes on the days of training or heat, and not jumping meals.

V2 genetic plan and medication: Compatible with the plan (hydration already recommended). • *suggestion (not applied)*

4. Caffeine: constant dose and time

Genetics: CYP1A2 *1F A/C; ADORA2A CC

Keeping the plan limit (up to 200 mg before midday) every day, including weekend: caffeine withdrawal gives headache 12–24 h after the last dose.

V2 genetic plan and medication: The same as the plan. • *suggestion (not applied)*

5. If migraine appears: prevention options

Genetics: Scores p29 and p32; MTHFR G/A

With medical diagnosis: riboflavin 400 mg/day (prevention dose, different from the 5–25 mg of the plan), coenzyme Q10 100–300 mg/day, magnesium (already taken) and melatonin 0.3–1 mg 30–60 min before bedtime (the dose of the plan; melatonin has positive trials in preventing migraine; decide with the doctor).

V2 genetic plan and medication: Coherent with the plan in melatonin (0.3–1 mg); only the dose of riboflavin would change, and only if there were diagnosed migraine. • *suggestion (not applied)*

6. If you need acute treatment

Genetics: 9p21 risk homozygous (Advanced Cardiovascular Report)

The triptans contract the arteries; with the 9p21 of risk, talk to the doctor and consider the gepants (ubrogepant), which are not vasoconstrictors. Avoid analgesics more than 10 days per month (cephaly due to medication abuse).

V2 genetic plan and medication: No conflict. • *suggestion (not applied)*

Avoiding or with caution

- Tetracyclines (doxycycline, minocycline) with isotretinoin: intracranial hypertension. *suggestion (not applied)*
- Nitrates (nitroglycerin, amyl nitrate) with tadalafil: absolute contraindication; nitrates are also a trigger for migraine. *suggestion (not applied)*
- Analgesics or triptans more than 10 days per month. *suggestion (not applied)*
- Dehydration and long fasts without water, especially with dapagliflozin and retatrutide. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. The monogenic forms (CACNA1A, ATP1A2, SCN1A, KCNK18, NOTCH3/CADASIL) are not read.
- Migraine has more than 120 loci of small effect; the value is in the scores, and even these explain little variance. The headache in rescues has very few loci.
- Without clinical data on headaches, the report describes predisposition, non-diagnosis.
- Migraine with aura does not have its own score: the loci of GWAS Catalog for aura do not indicate the risk allele.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001282 (PGS Catalog; snpnet; Migration (time-to-event); PGP000244, PMID 35324888 · PGS004797 (PGS Catalog; PRSmix; Migraine; PGP000604, PMID 38508198 · PGS001928 (PGS Catalog; Penalized regression (bigstatsr); Headaches for 3+ months; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Hautakangas H et al. Genome-wide analysis of 102,084 migraine cases identifies 123 risk loci and subtype-specific risk alleles. *Nat Genet* 2022.
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- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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